

A Statistical Analysis of Wildland and Prescribed Fires in the Changing California Climate

Masters Thesis by Sequoia Andrade
San Francisco State University

Outline

1. Introduction: Motivation, Existing Research, and Research Questions
2. Dataset: Creation and Exploratory Analysis
3. Inferential Analysis
4. Time Series Analysis
5. Regression Analysis
6. Discussion
7. Conclusion

Motivation

- Climate change-driven large, destructive wildfires becoming more frequent, especially in California [1,2]

- Prescribed burns are known to be effective in reducing wildfire intensity in some scenarios [3], yet:
  - Environmental trends indicate the window for prescribed burns is shrinking
  - Limited implementation due to controversy on risk, cost, and environmental impact

- Much of the existing body of research was performed in the previous century and did not account for climate change's impact on prescribed burn efficacy

Statistical analysis can be used to identify effective or ineffective prevention strategies, such as prescribed burns, but the data needed to study this question is **fragmented** and **complex**

Existing Research

Model	Variables	Dataset	Source
Generalized Linear Mixed Models (GLMM)	Tree mortality, fire area as RE	SNFDP	[13, 35]
Generalized Linear Model (GLM)	Mortality 3 years post fire	N/A	[35]
Difference of Proportion – Fischer's	Tree mortality, tree species, beetles, drought	Tree Mortality	[38]
Chi-Squared	Burn vs unburned, mortality between scare groupings	N/A	[23]
Cross-correlation	Wildfire vs climate variables	Multiple	[24,30,33,34]

Table 1: Existing research methods studying wildfires and their relations with other variables

Research Questions

1. Are prescribed burns effective at reducing the size of wildfires?

2. Are prescribed burns effective at reducing the size of wildfires while controlling for climate-relevant variables

3. How do these relationships change at different spatio-temporal scales?

We Investigate these questions with:

- Inferential Statistics - Mann-Whitney U-test and Chi-Squared test
- Monthly Time Series Analysis – Correlation and Cross-Correlation
- Regression Analysis – Linear, Non-Parametric, and Tree-Based Modeling

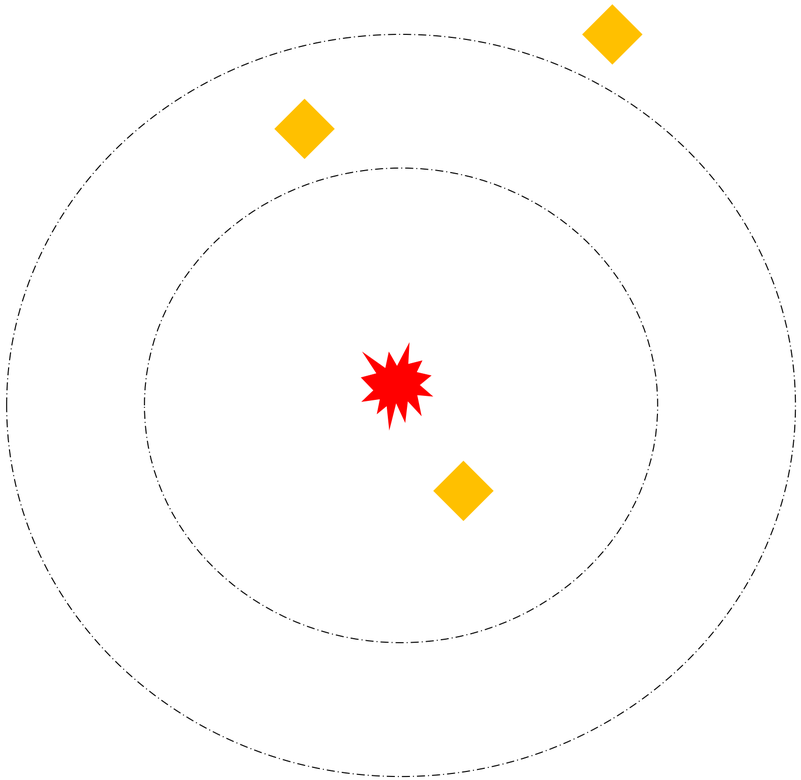
Data

Variable	Source	Description	Range	Units	Symbol
Acres Burned	CALfire[4]	Wildfire acres burned	0+	Log(Acres)	log_acres
Treatment Acres	CALfire[5]	Prescribed fire acres burned	0+	Acres	rx_acres
Precipitation	NOAA[6]	Monthly Precipitation averaged across the county	0+	Inches	pcp
Temperature	NOAA[6]	Average, minimum, and maximum temperature averaged across the county	0+	Fahrenheit	temp
Palmer Index	NOAA[6]	Drought severity indices	[-10, 10]	N/A	zndx, pdsi, phdi, pmdi
Windspeed	OpenMeteo [7]	Maximum windspeed on fire start date	0+	Km/h	max_wind

Table 2: Variables in the dataset, including descriptions, sources, and units

Dataset Creation

- Created by combining NOAA climate data and CALFire fire history data
- Data from 2041 wildfires spanning January 2013 to December 2022
- Columns:
    - wildfire acres burned, precipitation, temperature, wind speed, drought index
    - for each 3-36 month time frame, get prescribed fires in 5 - 100 km distance radius



Data Overview

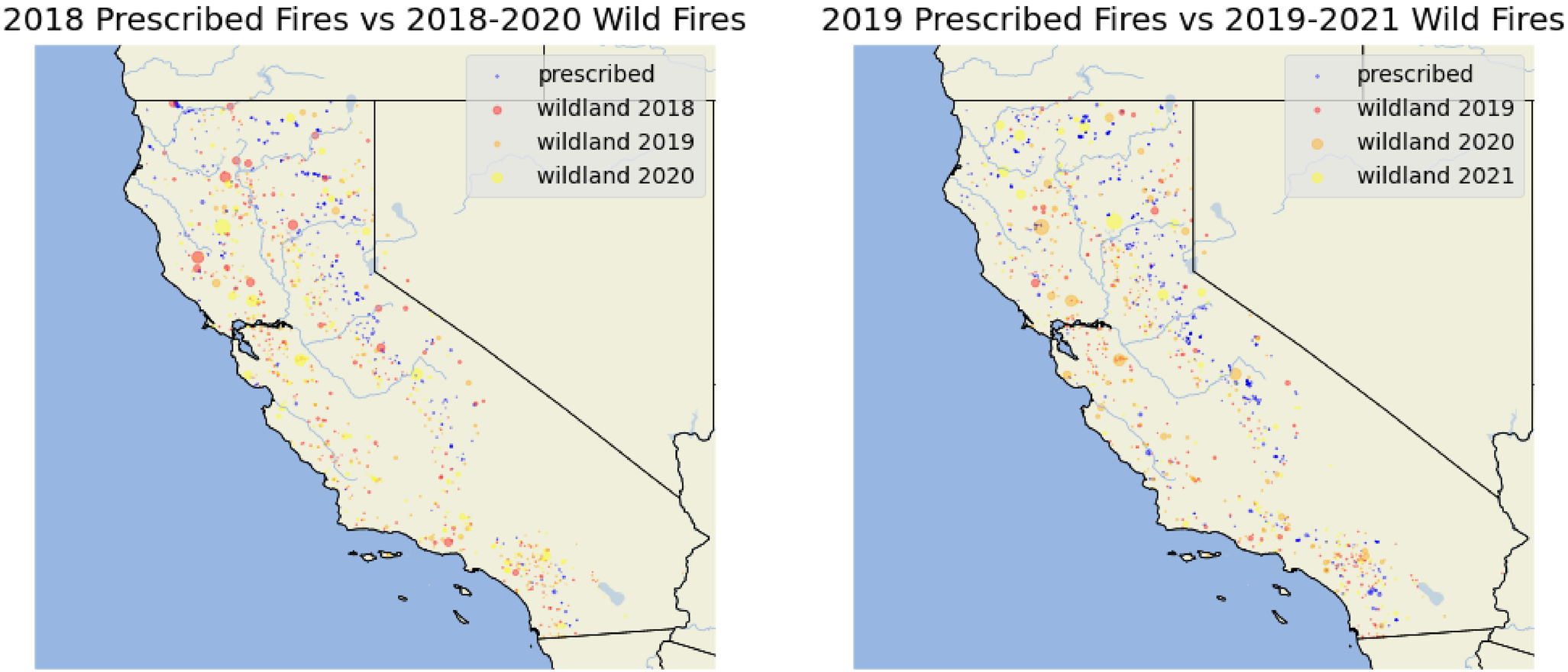


Figure 1: Visualization of prescribed fires for each year and wildfires in following years

Data Overview

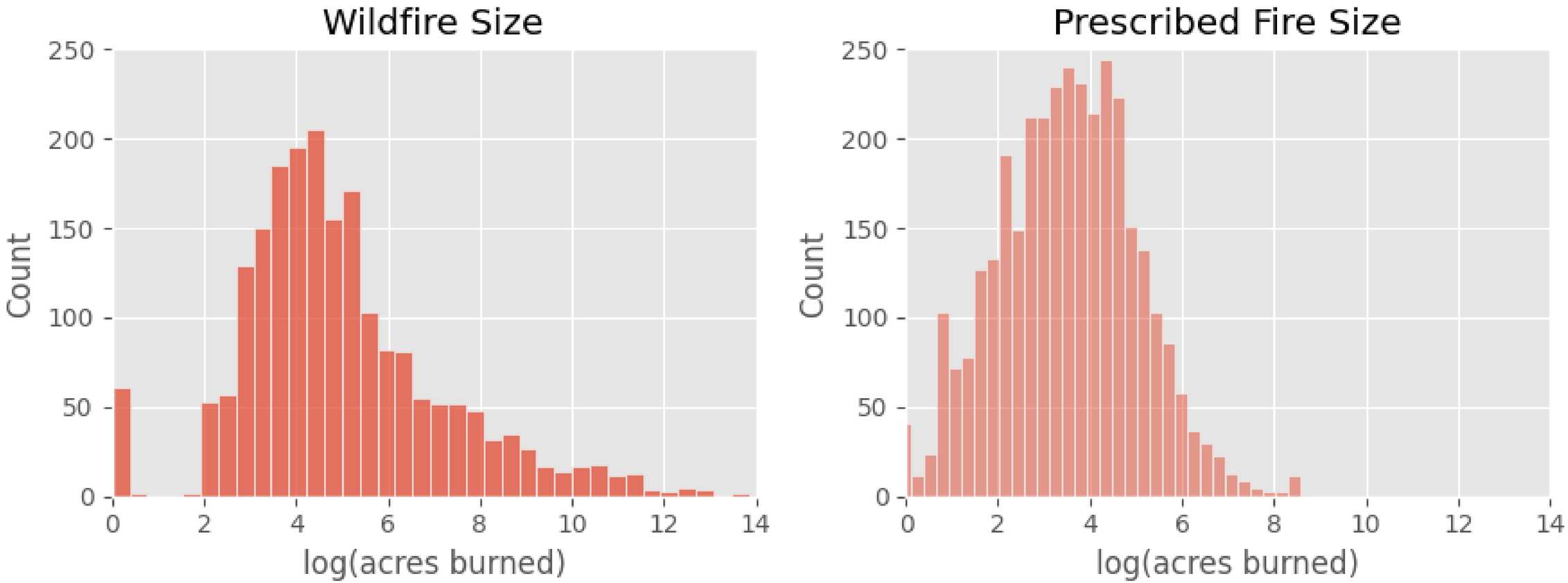
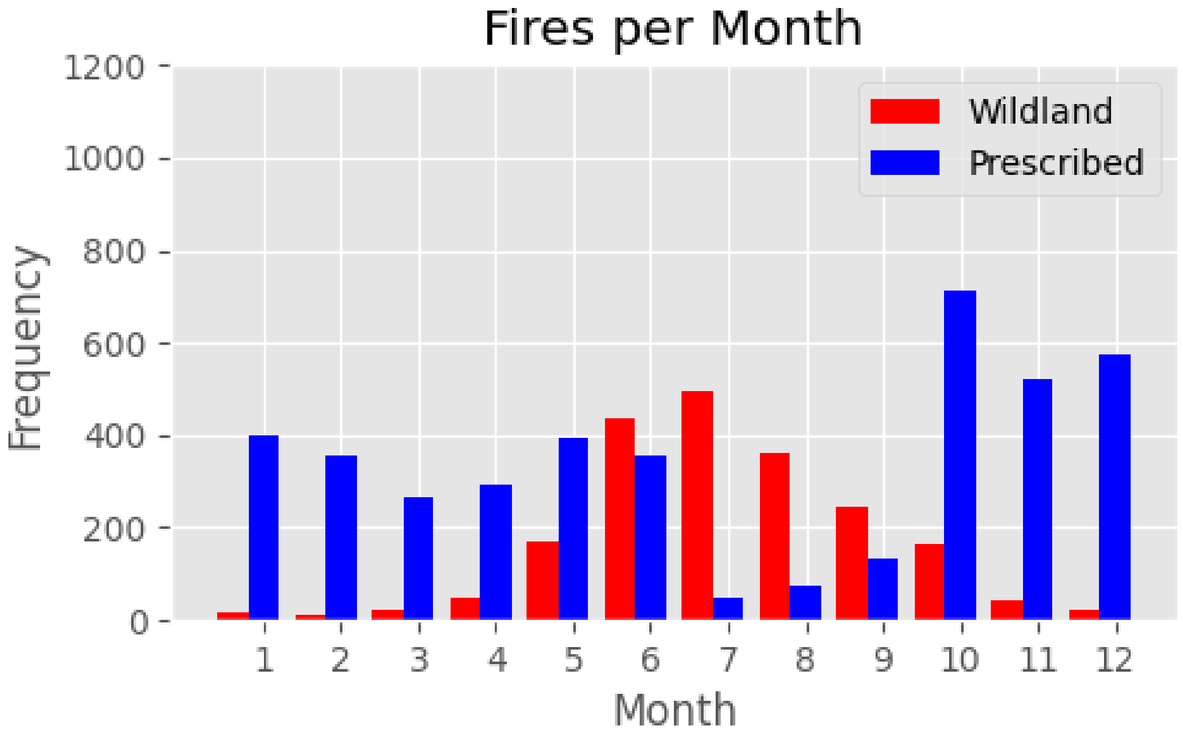


Figure 2: Distribution of wildfire and prescribed fire size

Data Overview



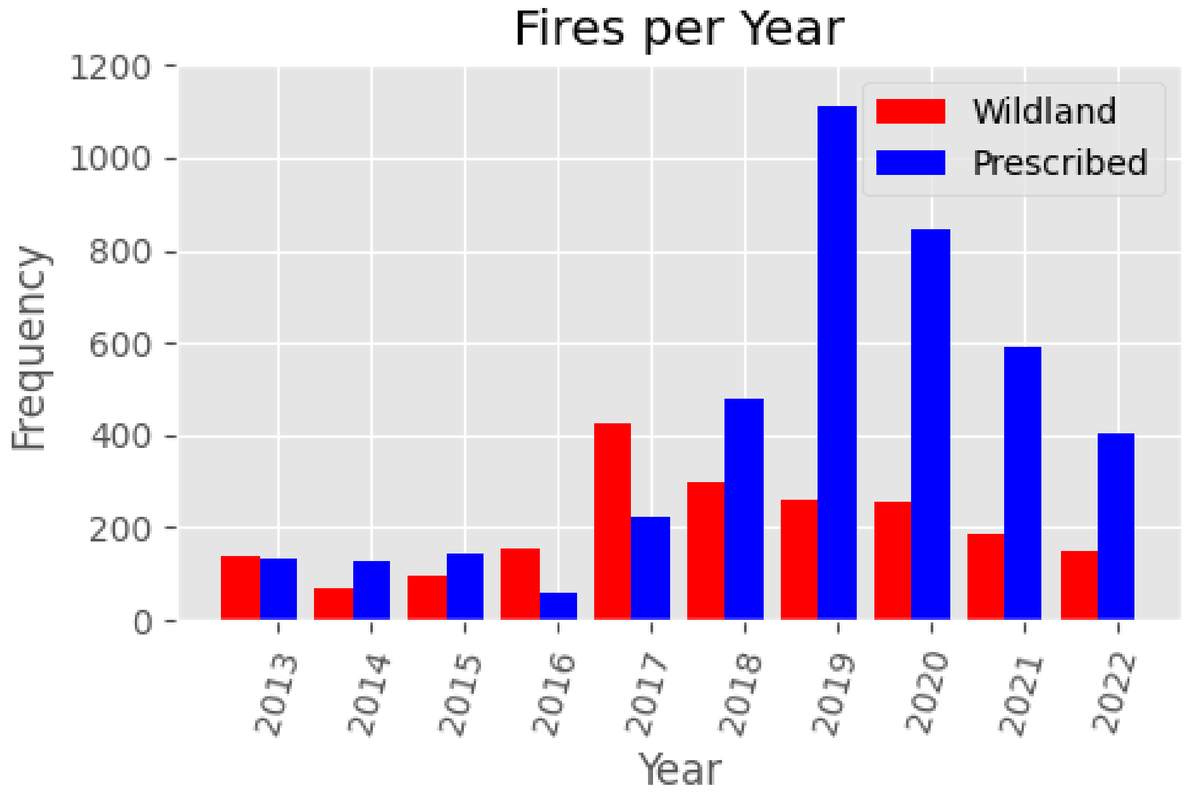


Figure 3: Distribution over years and months of wildfires and prescribed fires.

Methods

- **Inferential analysis:**
  - Difference of median test to examine if wildfires are smaller when there is a prescribed fire nearby
  - Chi-squared test to examine distribution of wildfires and prescribed fires over regions
- **Time series analysis:**
  - Correlation to identify which predictors are associated with acres burned
  - Cross-correlation to identify at what time lags predictors are associated with acres burned
- **Regression Analysis:**
  - Predict acres burned per wildfire incident from prescribed fires and climate variables

Inferential Analysis

Inferential statistics are applied to the data set to investigate:

1. If wildfire size varies across different climate and prescribed fire conditions
2. The distribution of wildfires and prescribed fires over regions

Two methods are used:

- **Mann-Whitney U-test** (Wilcoxon-signed rank test):

$$U = \min(U_1, U_2)$$

$$U_1 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2;\ U_2 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1$$

- **Chi-squared test:**

$$X^2 = \sum_{i=1}^{k} \frac{(n_i - \mu_i)^2}{\mu_i}$$

Inferential Analysis

Figure 4: Box plot of wildfire size between groups with and without prescribed burns within the specified distance

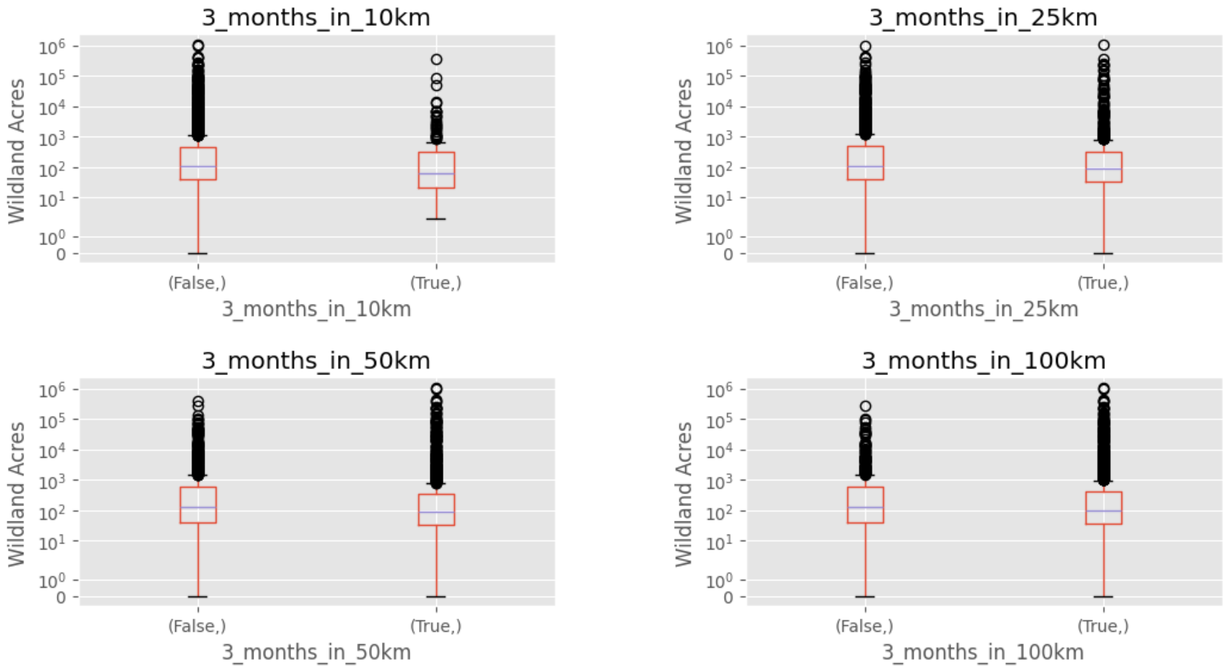


Table 3: Mann-Whitney U-test results, including rank biserial correlation (RBC) and common language effect size (CLES)

		U-val	alternative	p-val	RBC	CLES
prescribed_3_months_in_10km	MWU	116462.5	less	0.002	0.142	0.571
prescribed_3_months_in_25km	MWU	398860.5	less	0.015	0.062	0.531
prescribed_3_months_in_50km	MWU	470156.5	less	0.000	0.085	0.542
prescribed_3_months_in_100km	MWU	324331.0	less	0.044	0.054	0.527

Prescribed burn treatment is associated with fewer acres burned at 10 km, 25 km, 50 km, and 100km distance when completed 3 months before a fire

Inferential Analysis

Figure 5: Box plot of wildfire size between groups with and without prescribed burns within the specified distance

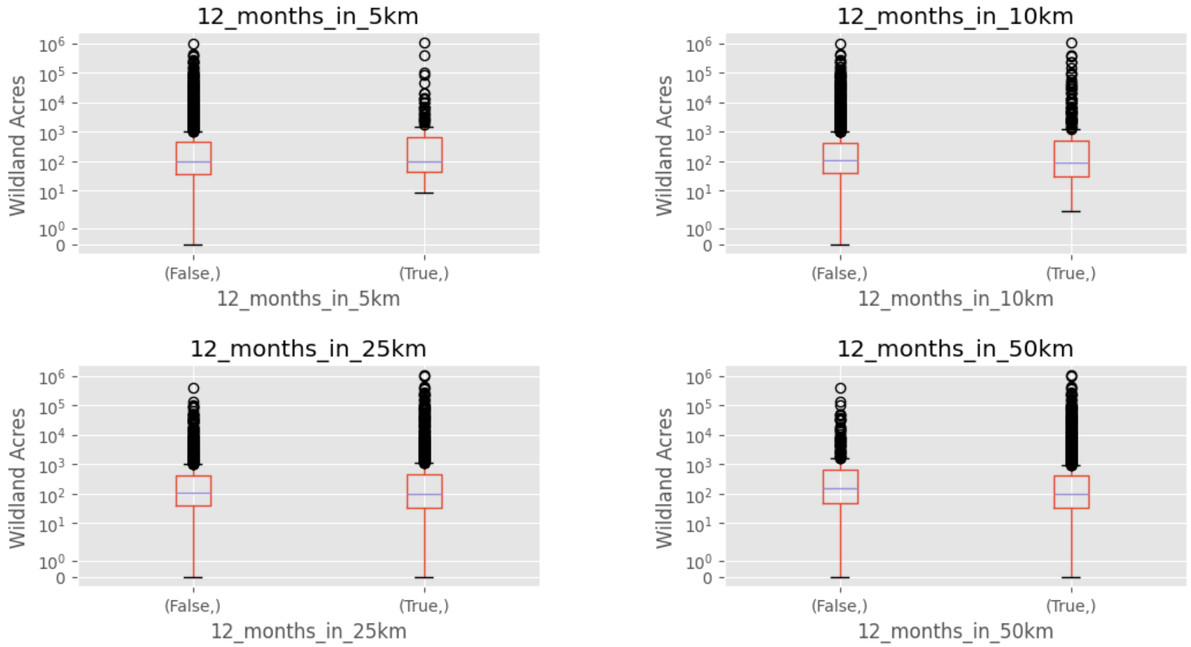


Table 4: Mann-Whitney U-test results, including rank biserial correlation (RBC) and common language effect size (CLES)

		U-val	alternative	p-val	RBC	CLES
prescribed_12_months_in_5km	MWU	136871.0	less	0.618	-0.015	0.492
prescribed_12_months_in_10km	MWU	298599.0	less	0.039	0.058	0.529
prescribed_12_months_in_25km	MWU	500004.5	less	0.194	0.022	0.511
prescribed_12_months_in_50km	MWU	214617.0	less	0.000	0.132	0.566

Prescribed burn treatment is associated with fewer acres burned at 10 km 50 km distances when completed 12 months before a fire

Inferential Analysis

Figure 6: Box plot of wildfire size between groups with and without climate variables in the specified ranges

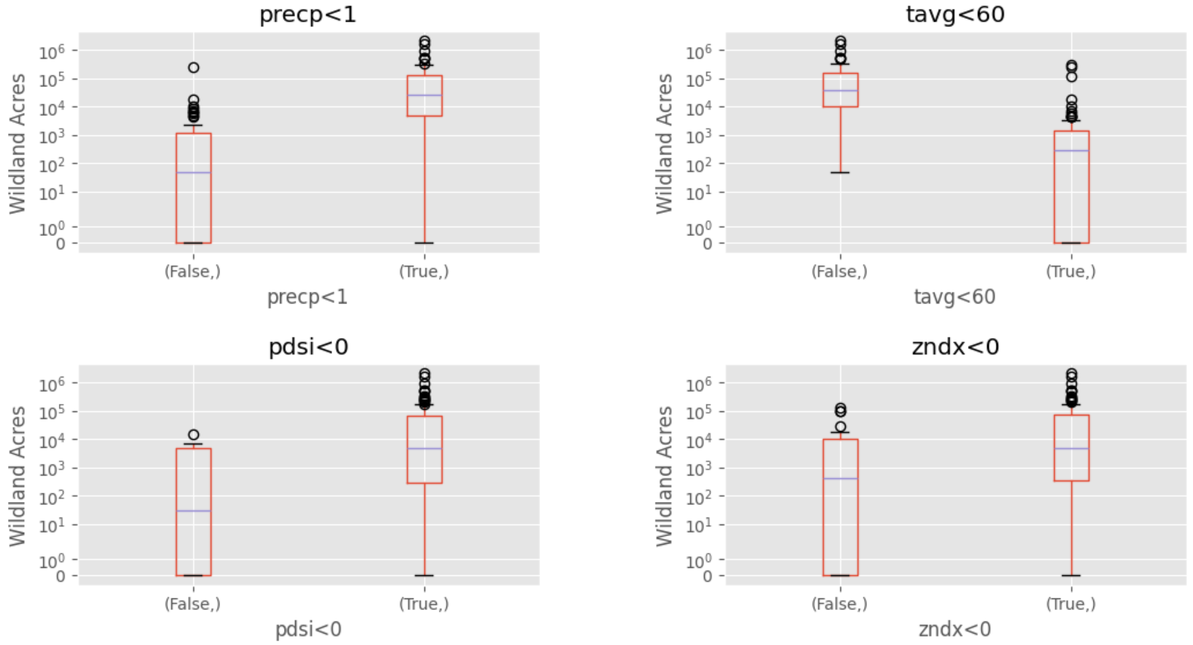
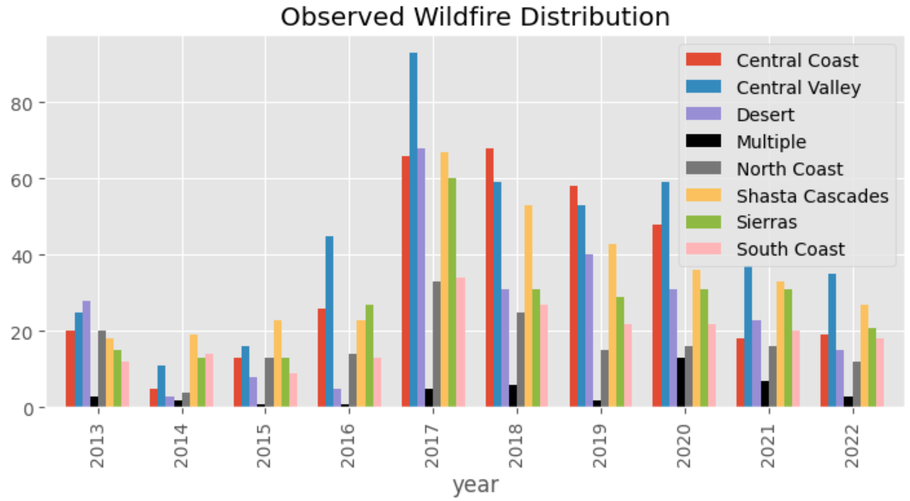


Table 5: Mann-Whitney U-test results, including rank biserial correlation (RBC) and common language effect size (CLES)

		U-val	alternative	p-val	RBC	CLES
precp<1	MWU	3117.5	two-sided	0.000	-0.772	0.886
tavg<60	MWU	274.5	two-sided	0.000	0.847	0.077
pdsi<0	MWU	1007.5	two-sided	0.008	-0.449	0.724
zndx<0	MWU	1654.0	two-sided	0.012	-0.317	0.659

Precipitation < 1 in, temperature > 60, and drought are associated with greater acres burned

Inferential Analysis



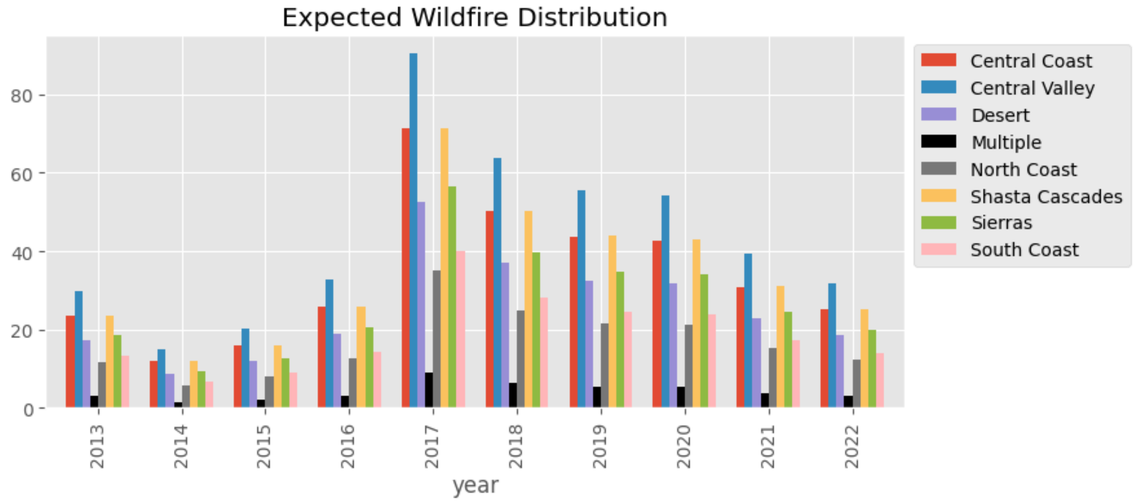


Table 6 : Chi-squared results for region and year fire frequency

X^2	Df	P-val
126.836	63	<0.001

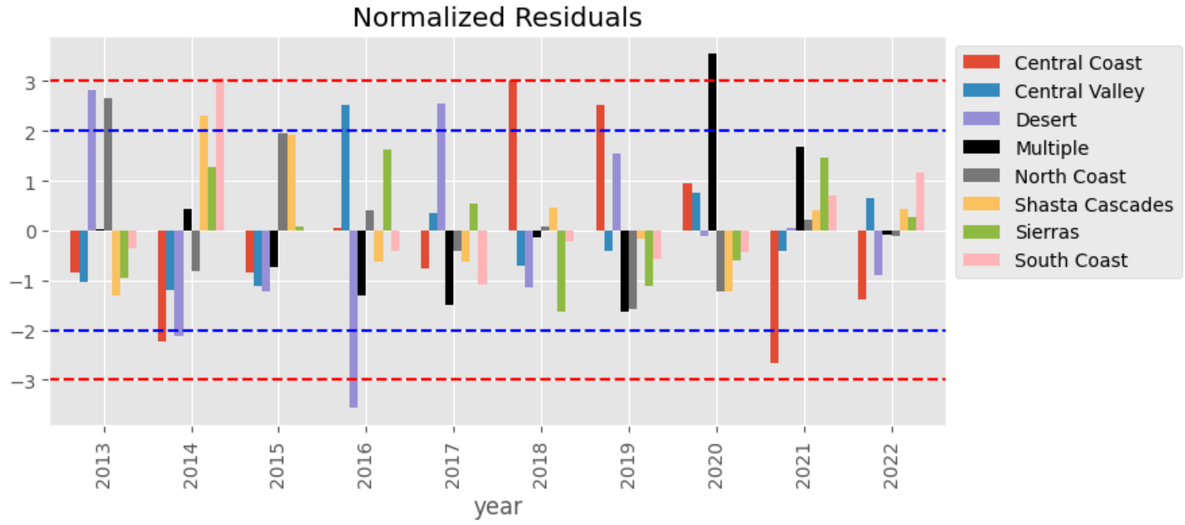
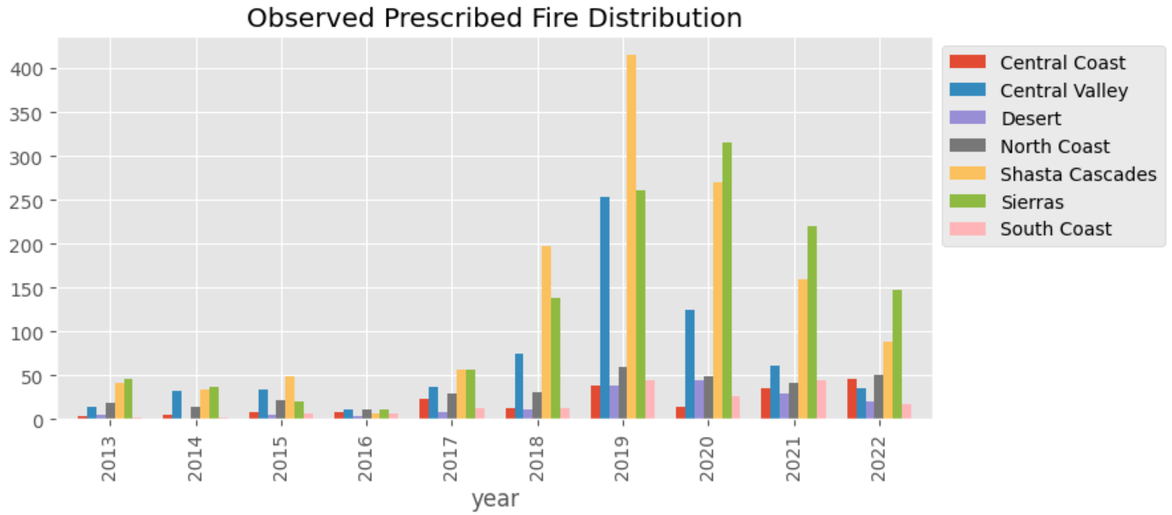


Figure 7: Observed, expected, and residuals for wildfire frequency between regions and years.

Inferential Analysis



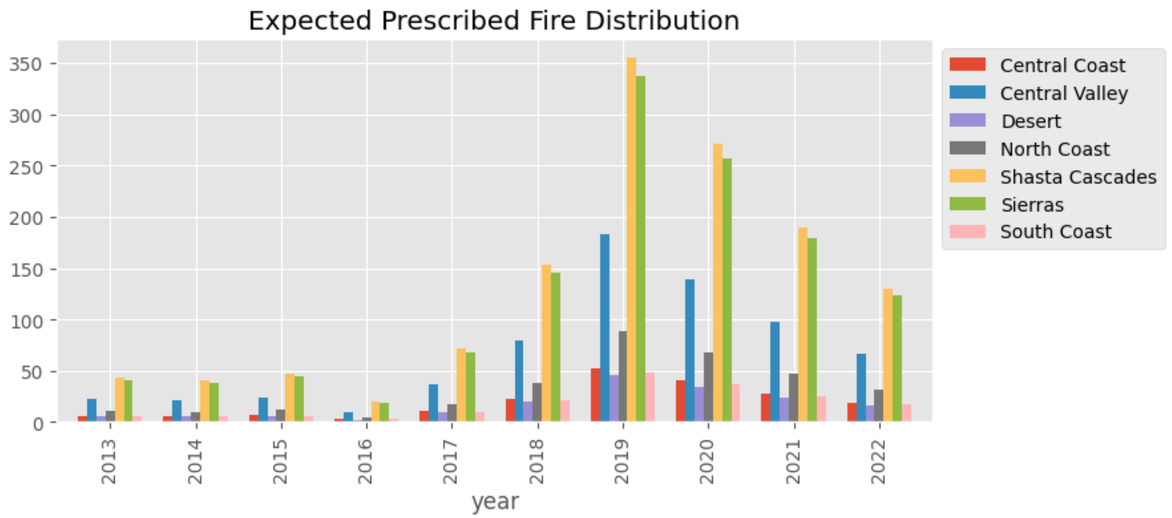


Table 7: Chi-squared results for region and year fire frequency

X^2	Df	P-val
382.960	54	<0.001

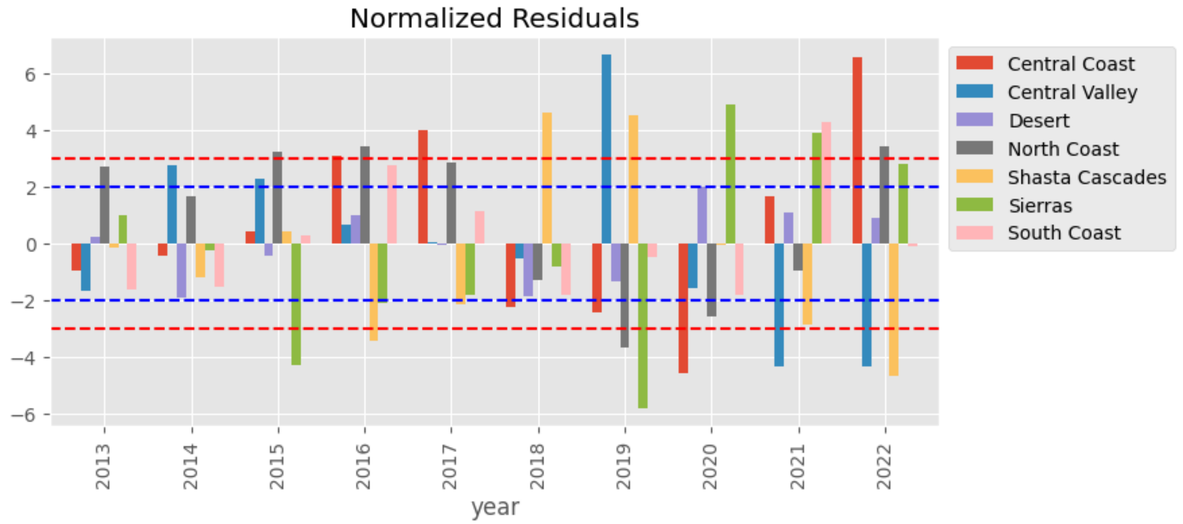


Figure 8: Observed, expected, and residuals for prescribed fire frequency between regions and years.

Inferential Analysis: Discussion

- Wildfires with a prescribed fire 10-100 km nearby within 3 months prior to ignition are smaller on average
- Wildfires with low precipitation, high temperature, and drought are larger on average
- Wildfire and prescribed fire frequency varies across regions over years

Limitations:

- This analysis did not consider temporal effects of climate on wildfire
- The analysis on prescribed burns also does not control for climate variables

Time Series Analysis

- Time series analysis is used to examine how trends in monthly wildfire acres burned is related to monthly climate variables and monthly prescribed burns
- Two methods are used:
    1. Correlation:

$$R = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

    2. Cross-correlation:

$$R_d = \frac{\sum_{i=1}^{n-d}(x_{i+d} - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n-d}(x_{i-d} - \bar{x})^2}\sqrt{\sum_{i=1}^{n-d}(y_i - \bar{y})^2}}$$

Time Series Analysis: Correlation

Wildfire acres burned have the strongest correlation with climate variables

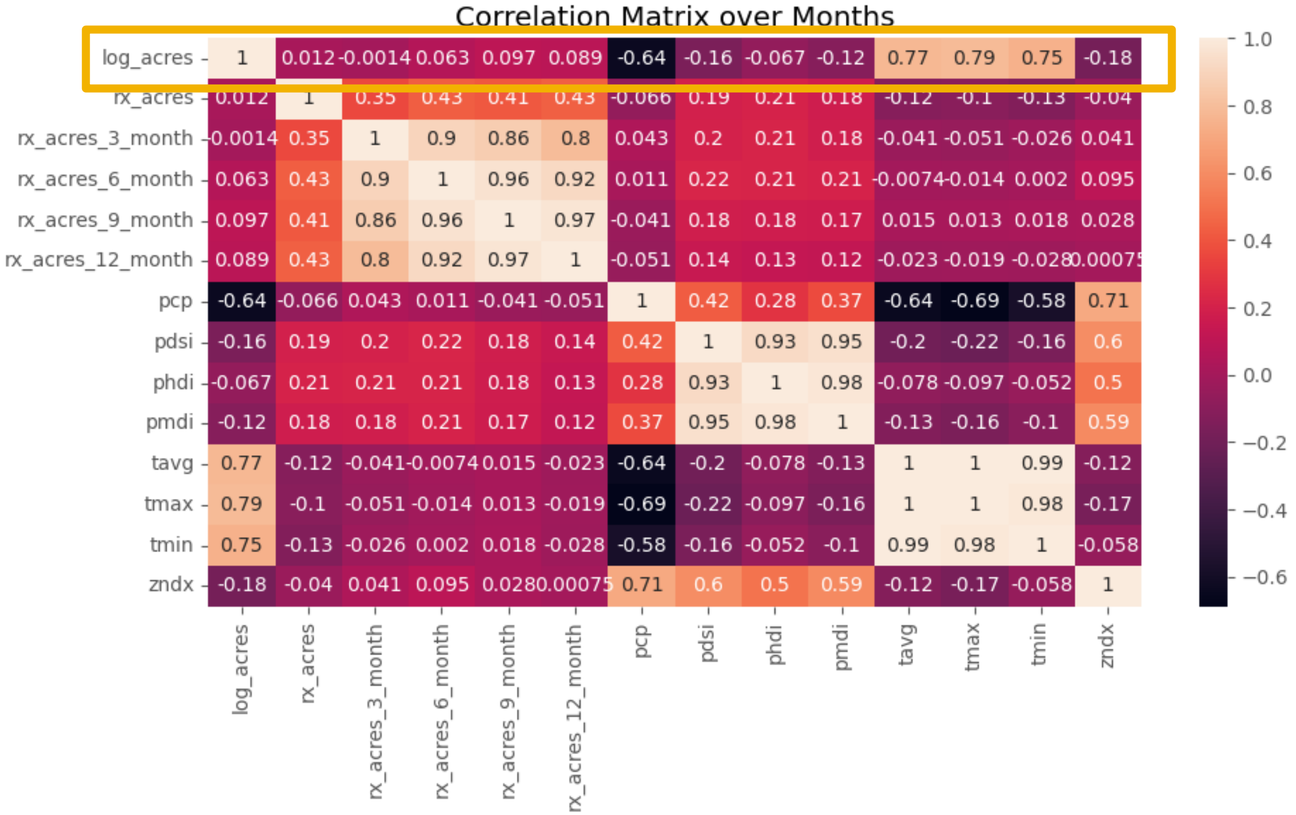


Figure 9: Correlation matrix for predictors and wildfire size

- Climate variables are correlated with each other
- Prescribed burn acres has no association with wildfire acres on this monthly scale
- Temperature and precipitation have the strongest relationships with wildfire acres burned

Time Series Analysis: Cross-Correlation

Temperature and drought index have the strongest relation to wildfires in the same month, while prescribed burns are strongest related at two months

Predictor	Best + Lag	Max R	Best - Lag	Min R
Rx acres	20	0.150	2	-0.077
Pcp	18	0.492	0	-0.636
Pdsi	16	0.222	0	-0.160
Phdi	15	0.227	0	-0.067
Pmdi	16	0.251	0	-0.120
Tavg	0	0.773	6	-0.702
Tmax	0	**0.788**	6	**-0.703**
Tmin	0	0.746	6	-0.694
zndx	19	0.139	0	-0.185

Table 8: Cross-correlation values for predictors and wildfire size

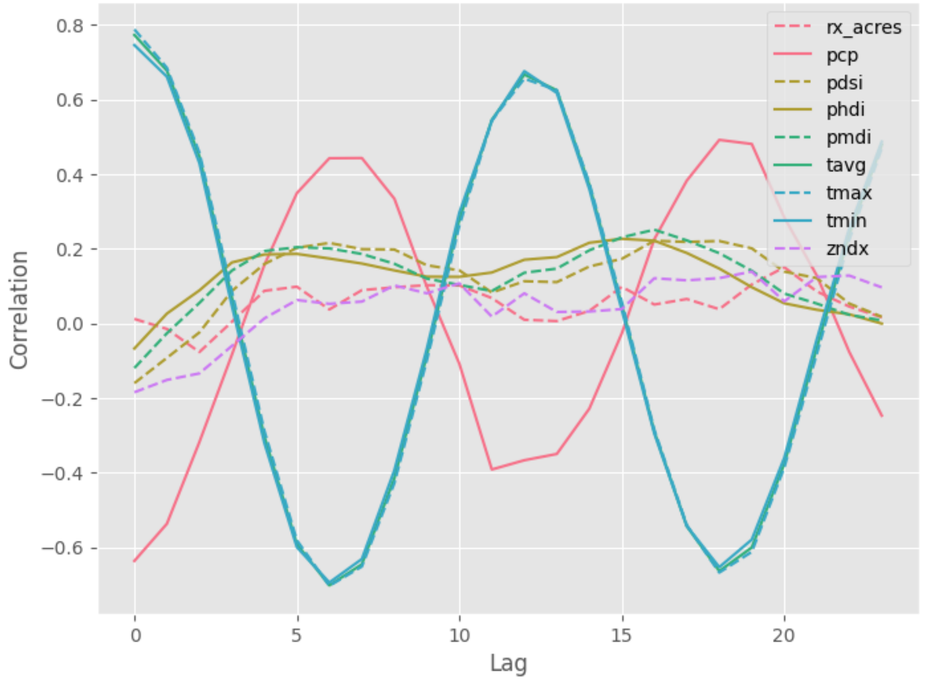


Figure 10: Cross-correlations for predictors and wildfire size

Time Series Analysis: Discussion

- Prescribed burns have the strongest negative effect on wildfire acres 2 months prior
- Temperature has the strongest correlation to wildfire acres burned
- The correlation of wildfire acres with temperature and drought has the greatest magnitude in the same month

Limitations:

- This analysis only considers pairwise relationships
- Relies on monthly totals

Regression Analysis

- Four model families:
  1. Linear regression
  2. Mixed-Effects Model
  3. Non-parametric: KNN, SVM
  4. Tree-based: Random Forest, AdaBoost
- Target Y: Log incident acres burned
- Input X: region, latitude, longitude, drought index, maximum temperature, precipitation, maximum wind, and prescribed burn variables

- Data split:
  - Split 80% training and validation, 20 % test
  - Trained and fine-tuned using 5-fold cross validation on the 80 % split
- Performance measured with MSE:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y_i})^2$$

Regression Analysis: Data Preprocessing

- Removed all wildfires with 0 acres
- Remove correlated predictors for linear models using Variation Inflation Factor (VIF)
- Standard Scaling:

$$x_{ij}^{*} = \frac{x_{ij} - \bar{x}_i}{s_i}$$

Predictor	VIF
precipitation	1.277
log RX 3 months 50km	2.121
max temp	13.582
max wind	12.686

Table 9: VIF scores for final linear model features

Regression Analysis: Linear Model

Model: $\hat{Y} = \hat{\alpha} + \hat{\beta}\boldsymbol{X}$

- $R^2 = 0.885$
- $Train\ MSE = 3.476$
- $Test\ MSE = 4.535$
- $F = 747.8, \mathrm{p} < 0.001$

Prescribed burns are related to decreases in wildfire size

Decreased precipitation is related to increases in wildfire size

Region has an effect on wildfire size

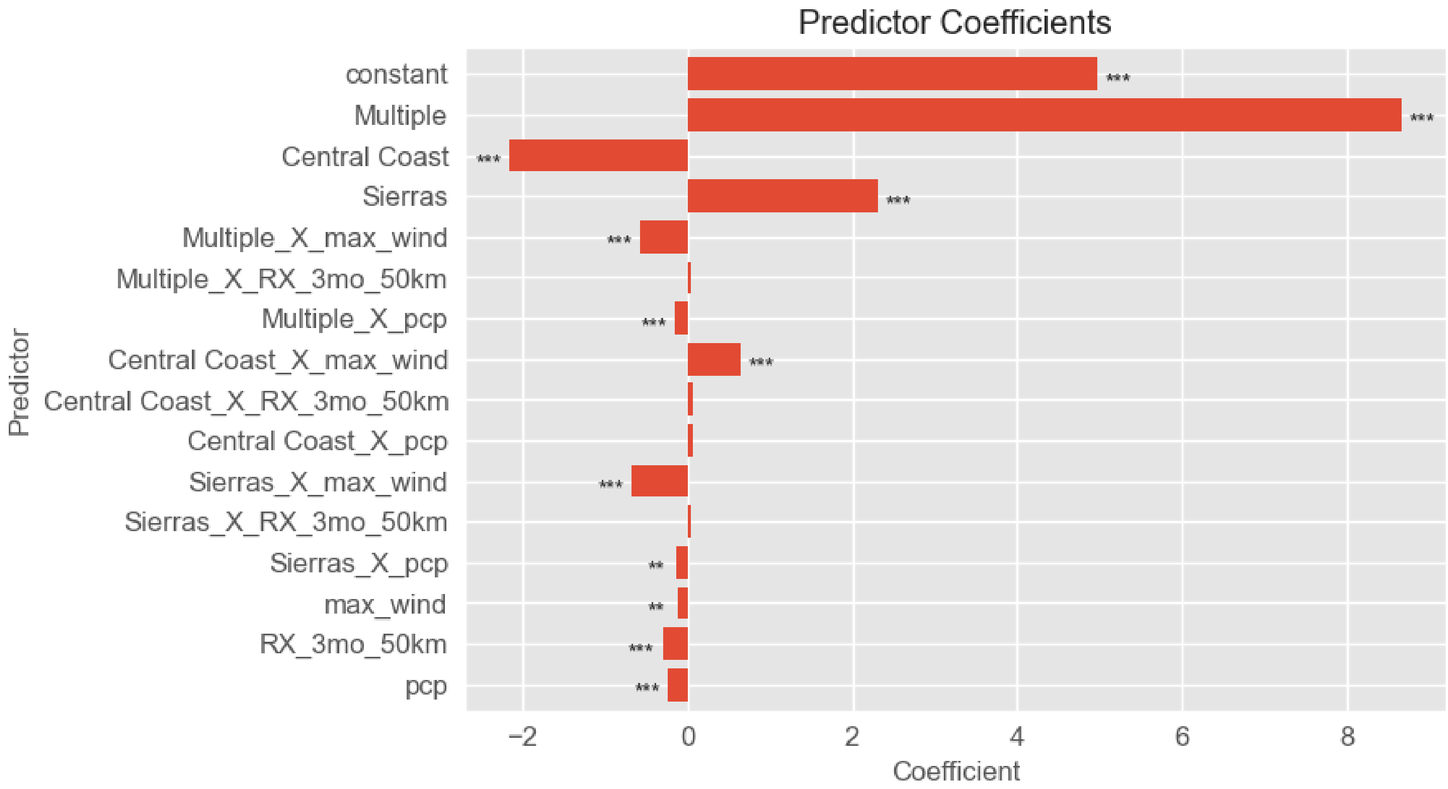


Figure 11: Regression coefficient results, with significant predictors notated with "*"

Regression Analysis: Linear Model

Figure 12: Residuals and partial regression for precipitation

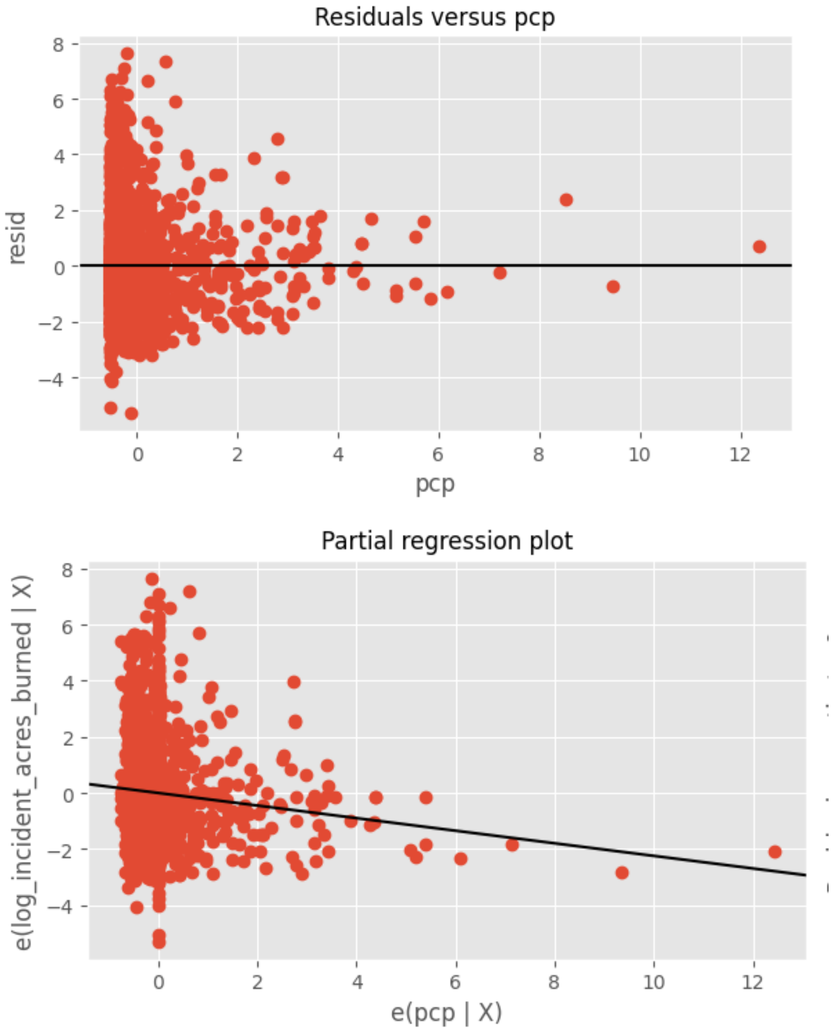
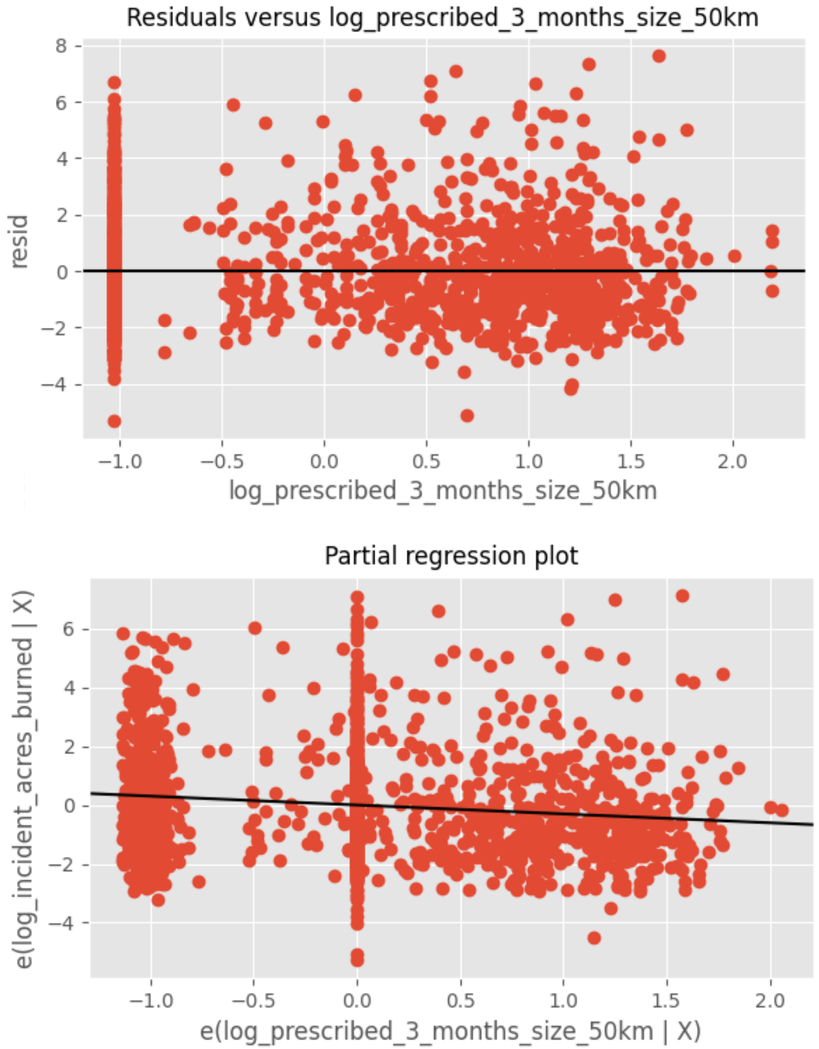


Figure 13: Residuals and partial regression for prescribed burns



Regression Analysis: Mixed-Effects Model

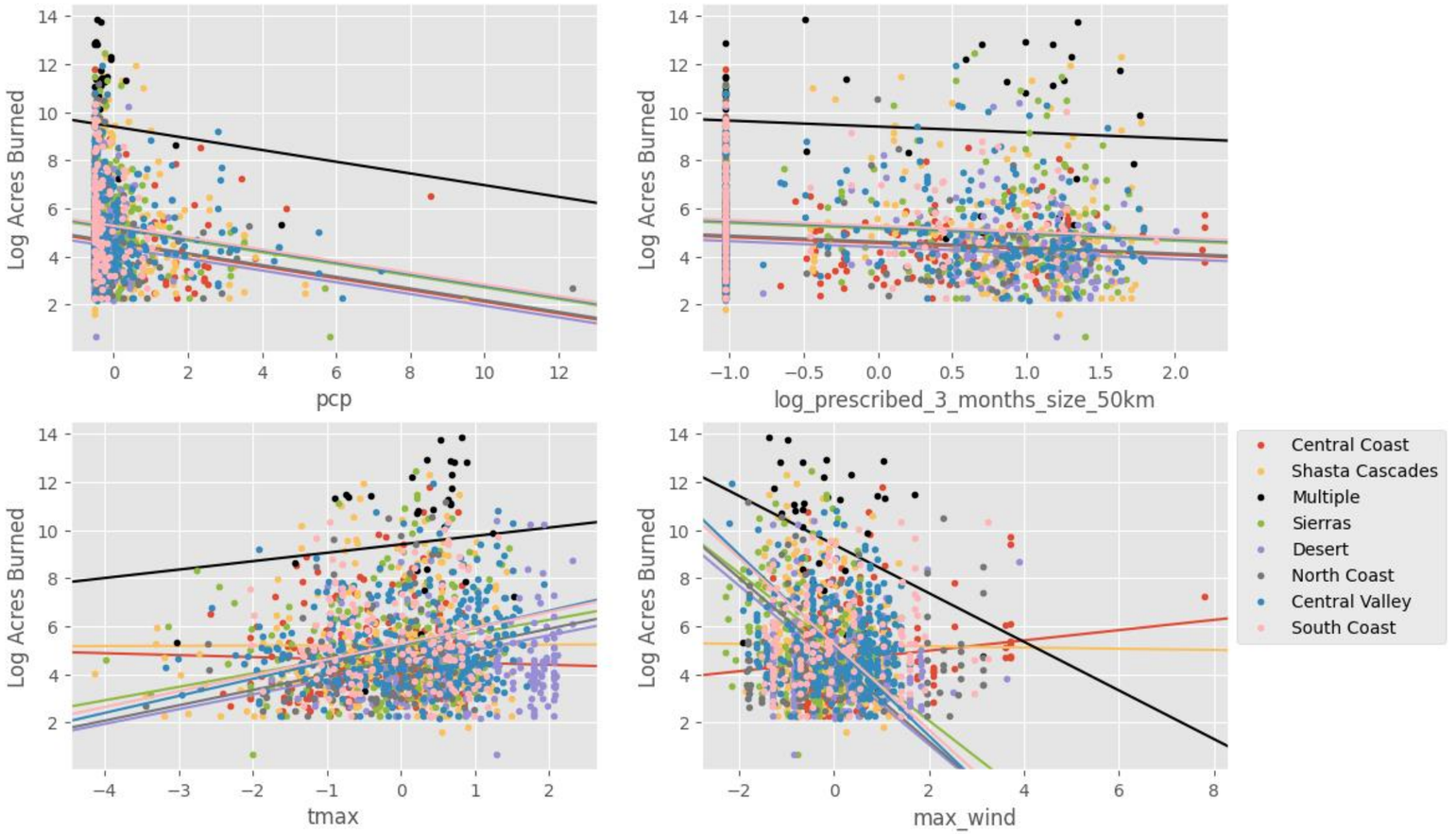
- Model: $Y_i = \beta_0 + \beta_1\, x_1 + \ \ldots + \beta_p\, x_p + \gamma_0 + \gamma_k x_k + \cdots + \gamma_l x_l + \epsilon_i$
- Fixed Effect: precipitation, log RX 3 mo 50km
- Random Intercept: region
- Random Slope: max temperature, max wind

Table 10: Linear mixed-effects regression results

	Coefficient	Std Error	Z	P(>|z|)
***intercept	4.796	0.311	15.397	<0.001
***pcp	-0.243	0.047	-5.140	<0.001
***log_RX	-0.248	0.051	-4.894	<0.001
Region Var	2.872	0.840	-	-
Region x tmax Cov	0.202	-	-	-
Tmax Var	0.025	-	-	-
Region x max wind Cov	-0.593	0.207	-	-
Tmax x max wind Cov	-0.067	0.011	-	-
Max_wind Var	0.185	0.061	-	-

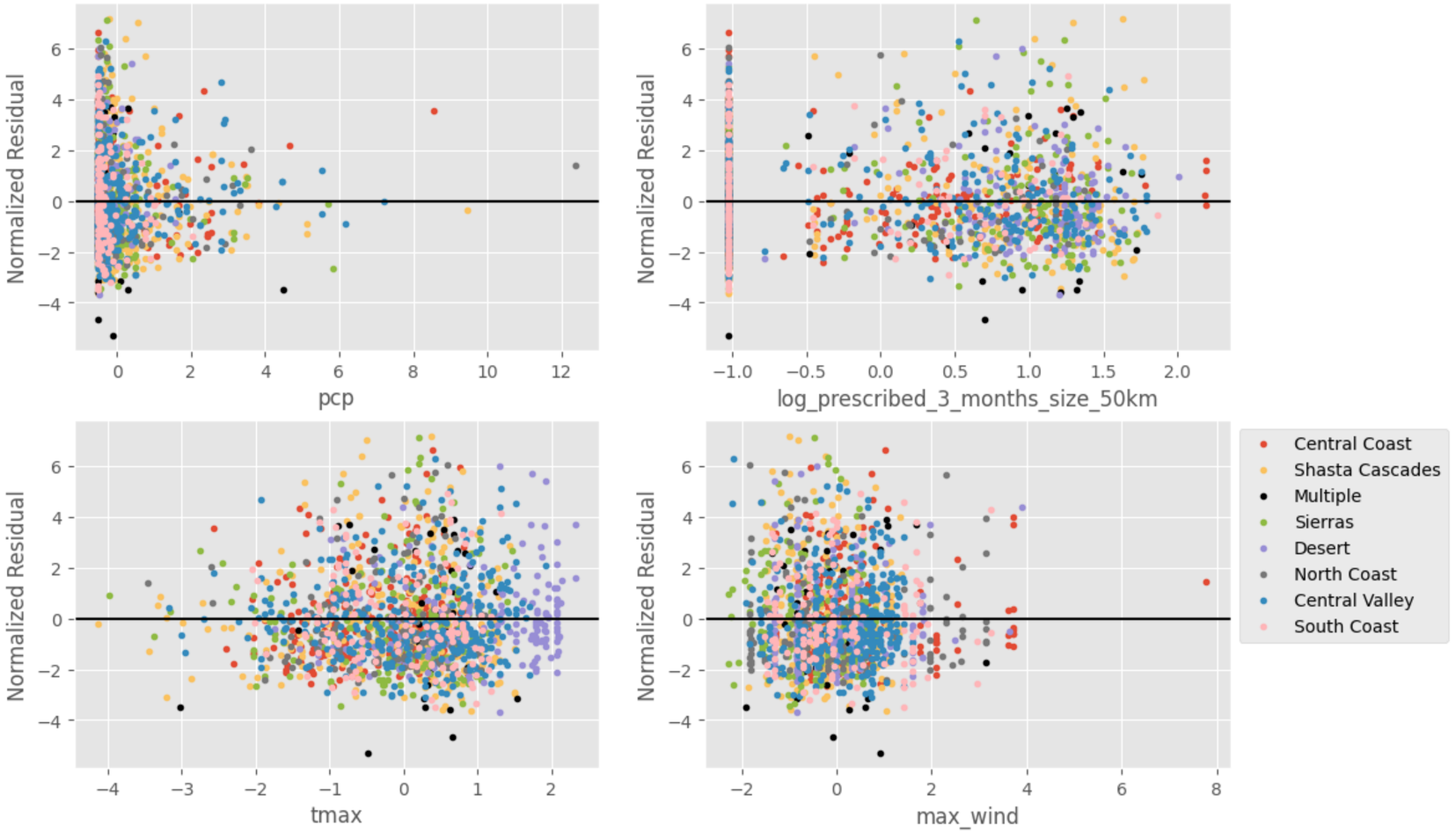
Regression Analysis: Mixed-Effects Model

Figure 16: Linear mixed-effects model partial regression plots



Regression Analysis: Mixed-Effects Model

Figure 15: Linear mixed-effects model residuals



Regression Analysis: Non-parametric Models

KNN: Uniform weights, k=15

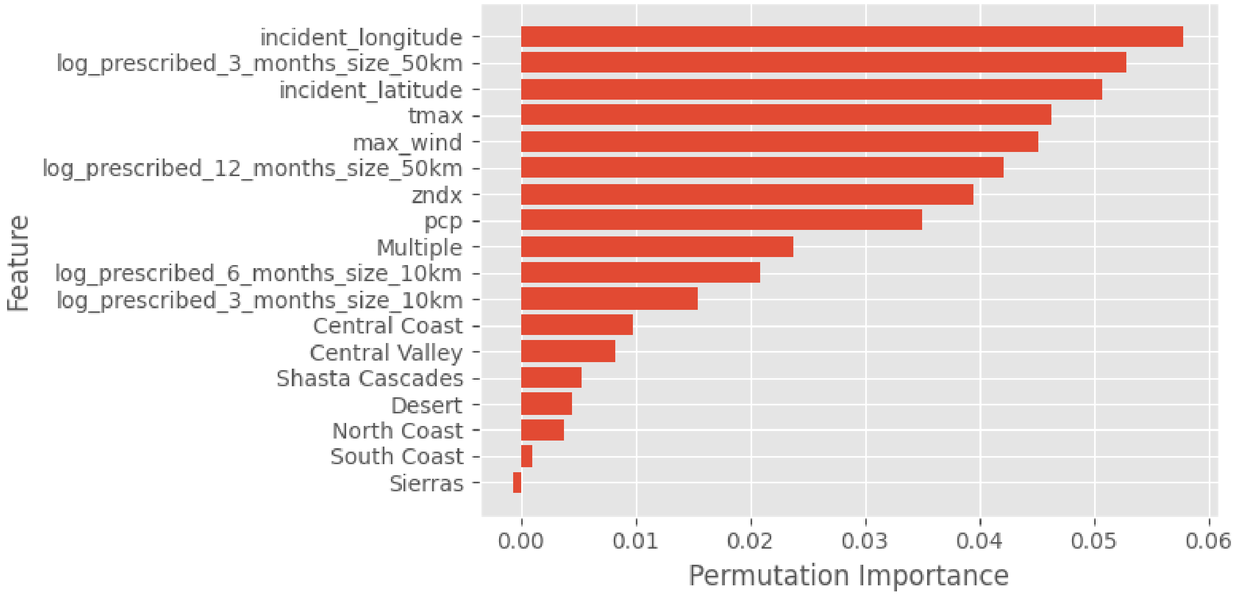


Figure 16: KNN Feature Importance

SVM: Linear kernel, C=1

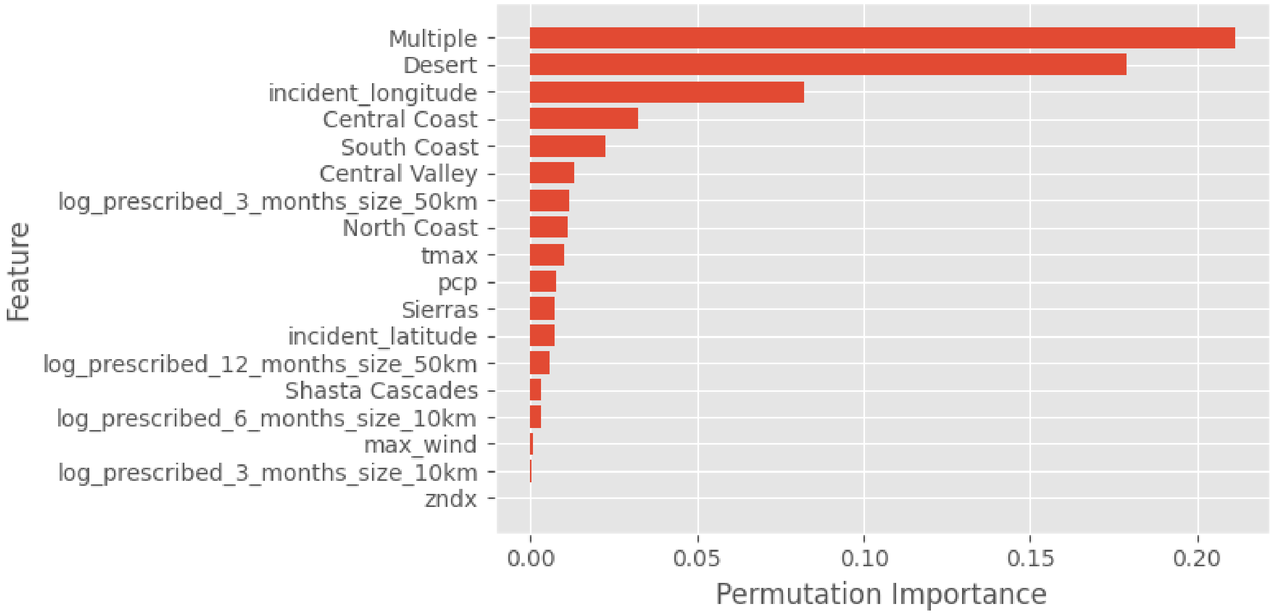


Figure 17: SVM

Regression Analysis: Tree Models

Random Forest: 1000 estimators, sqrt number of features

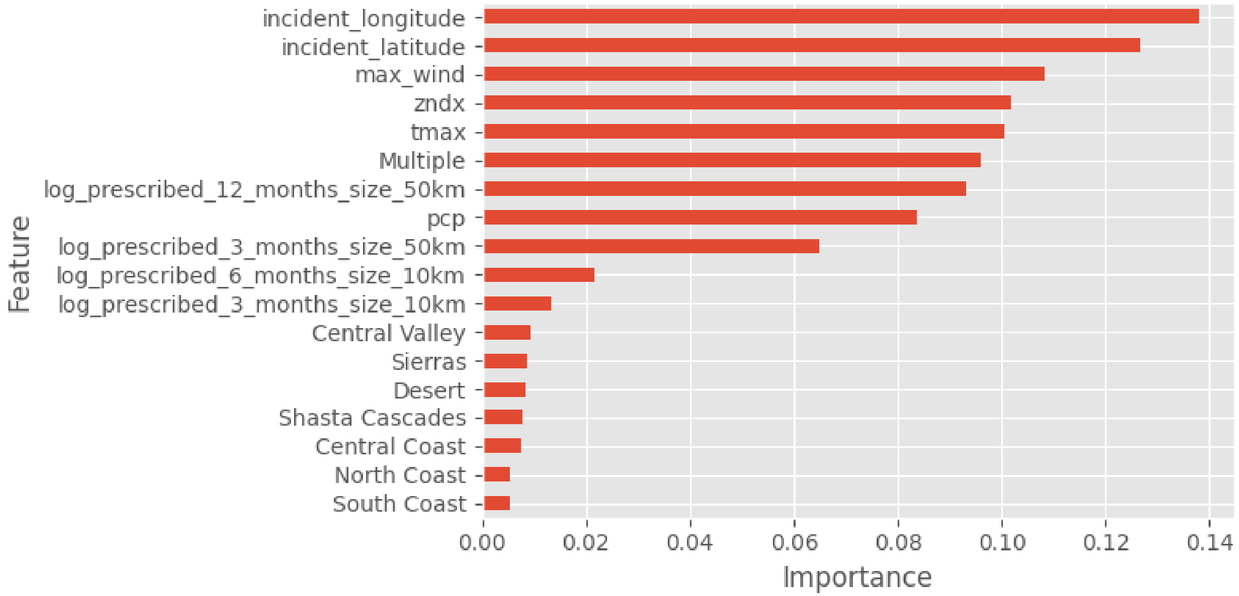


Figure 18: Random Forest Feature Importance

AdaBoost: 50 estimators

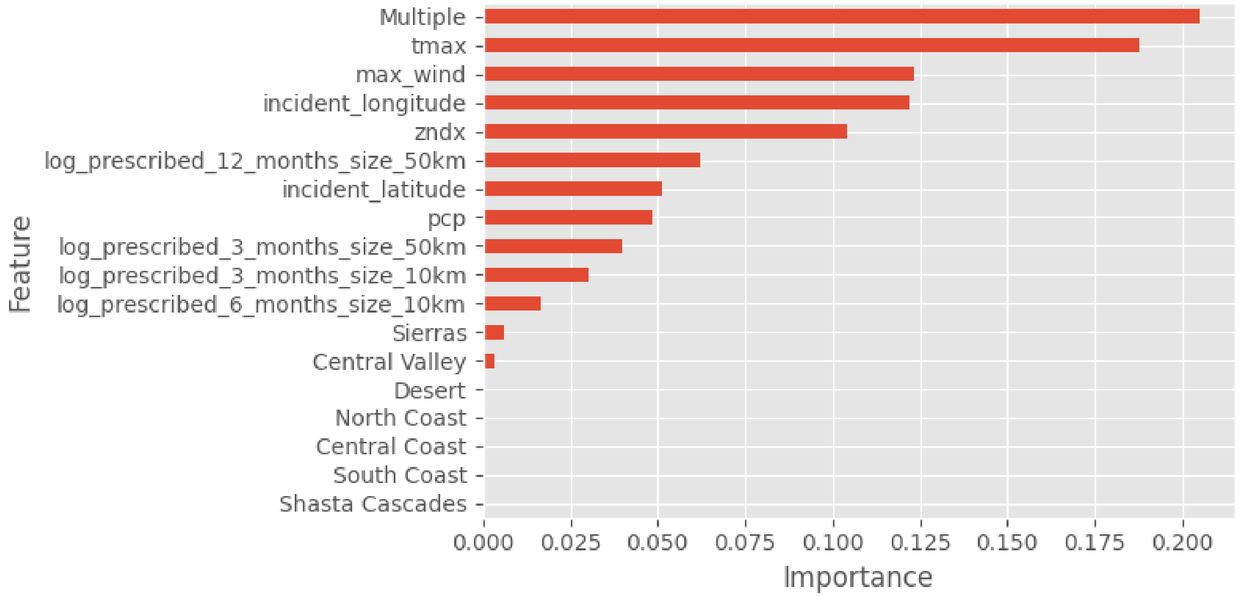


Figure 19: AdaBoost Feature Importance

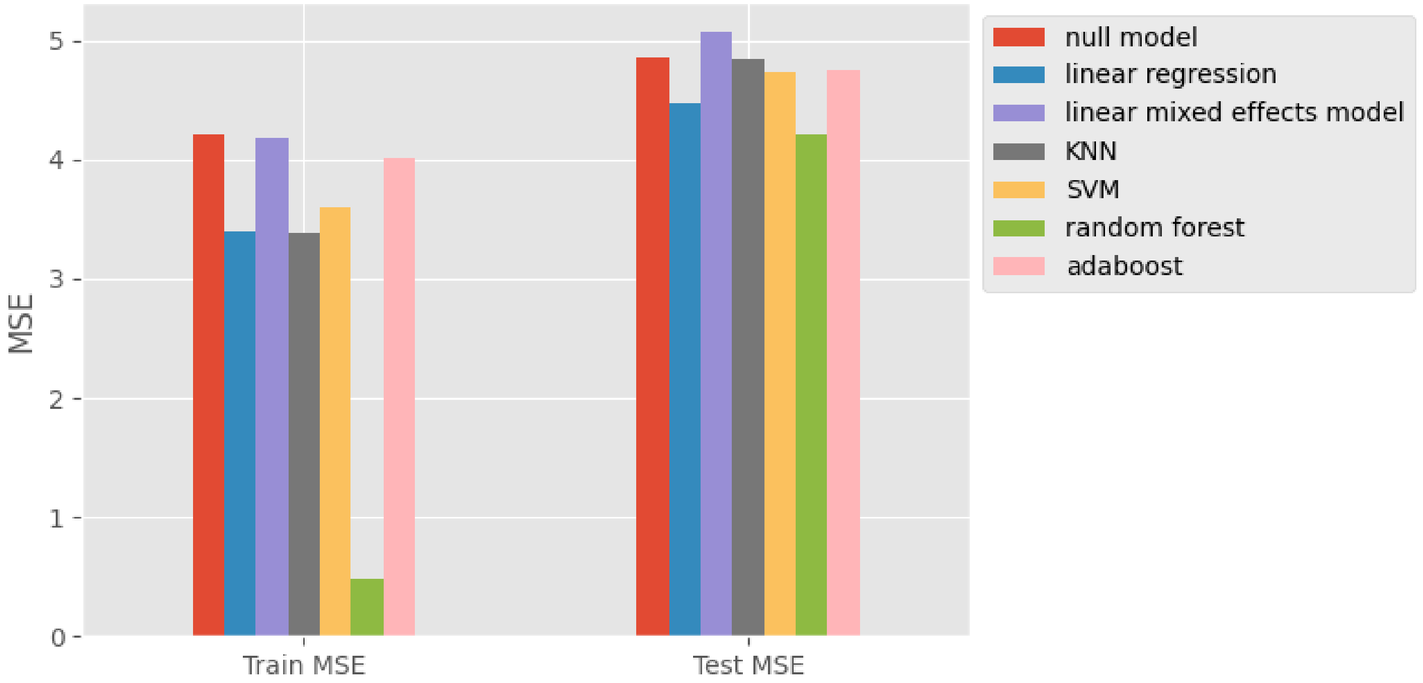
Regression Analysis: Model Comparison

- Models generalize well and do better on the test set
- Random forest has overfitting problem but still performs best

Table 11: Model train and test errors

Model	Train MSE	Test MSE
Null	4.209	4.857
Linear	3.396	4.467
Linear Mixed	4.187	5.066
KNN	3.380	4.847
SVM	3.594	4.734
Random Forest	**0.484**	**4.216**
AdaBoost	4.006	4.749

Figure 20: Plot of model performance metrics



Regression Analysis: Comparison

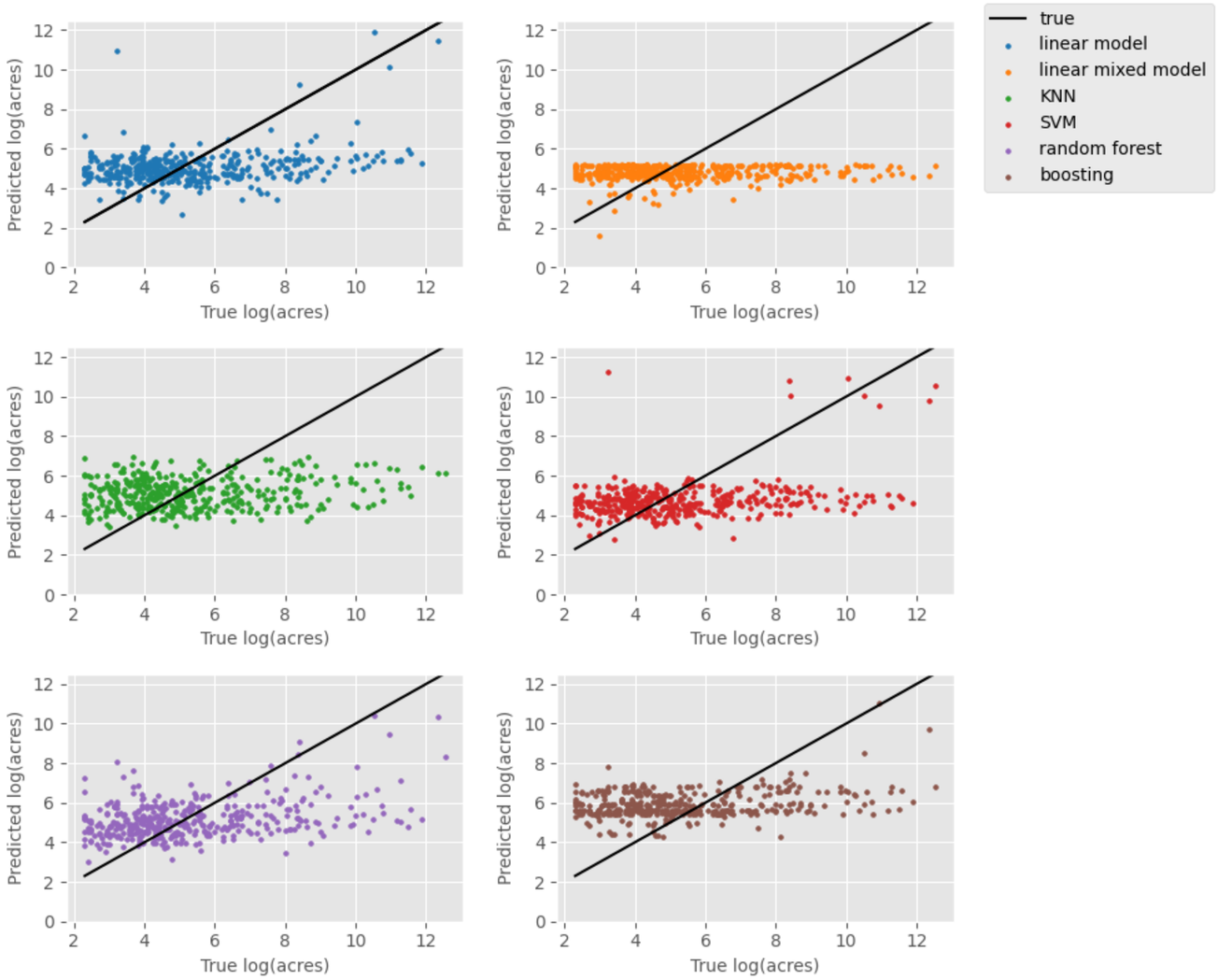


Figure 21: True vs predicted plot for each model

- The models all struggle with predicting extremely small and large fires
- The random forest performs the best with a few fires predicted well across the range
- The mixed model has little variation

Regression Analysis: Discussion

- Models generally agree that both climate variables are prescribed fires are important for wildfire size
- Precipitation, lower maximum temperature, wind speed, and prescribed fires are negatively related to wildfire size
- Prescribed fires do have a mitigating effect even when controlling for climate variables

Limitations:

- Variable distributions are not ideal for linear models
- Null model has low MSE
- Does not explicitly include spatial relationships

Recall: Research Questions

1. Are prescribed burns effective at reducing the size of wildfires?

2. Are prescribed burns effective at reducing the size of wildfires while controlling for climate-relevant variables

3. How do these relationships change at different spatio-temporal scales?

Results indicate:
- Increases in prescribed burn acreage are related to decreases in wildfire size
- The effect of prescribed burns is significant when controlling for precipitation
- The time and location of prescribed burns impacts efficacy

Discussion

- Increased prescribed burns are associated with decreased wildfire size for certain cases
- When controlling for climate variables, prescribed burns are still a significant predictor of wildfire size
- Prescribed burns have the greatest effect when performed within 3 months and 50 km of wildfires when controlling for precipitation and wind

Limitations:

- Vegetation type is not considered in this model
- Models use binning for handing spatial and temporal differences – an integrated model could be better

Conclusions

- Prescribed burns do have a mitigating effect on wildfire size, seen in both inferential and statistical analysis
- Prescribed burns tend to have the greatest impact when performed 2-3 months prior to fires, evident in all three analyses
- Though climate variables are strongly associated with wildfire size, prescribed burns still have an impact
- Policy makers and forest managers should consider prescribed burns in spring to have the greatest impact

References

See thesis paper



San Francisco State University